Lecture 8: Derived distributions

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Notice for Mid-term Exam

清華大學 Tsinghua University

- Date: April 20, 2021 (Week 9)
- Time: 9:50-11:25 (GMT+08:00, Beijing time) (In-class)
- Location: Lecture hall, 3-205, Rohm Building
- Coverage: Lecture 1 to 7
- Additional information:
  - Closed-book exam, with one **PAGE** of A4 “cheat sheet” available
  - Calculator **NOT** necessary
- Account for 25% in final grade
- Please get well-prepared! Good luck!

- If you cannot take the exam offline at the lecture hall, please fill the survey on `https://www.wjx.cn/vj/w14UQE9.aspx` **today**!

Outline

1. **Review**
2. **Derived distributions**
3. **Recitation**
4. **Summary**

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Review

- Joint PDF

$$\mathbf{P}((X, Y) \in S) = \iint_S f_{X,Y}(x, y)\, dx\, dy$$

- Conditioning

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \qquad \text{if } f_Y(y) > 0$$

- The continuous Bayes' rule

$$f_{X|Y}(x \mid y) = \frac{f_X(x) f_{Y|X}(y \mid x)}{\int_{-\infty}^{\infty} f_X(t) f_{Y|X}(y \mid t)\, dt}$$

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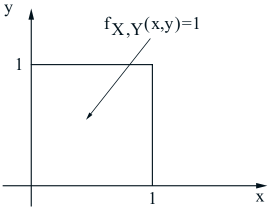
What is a derived distribution

- There is a PDF of one or more RVs with known probability law, e.g.:



- Obtaining the PDF for a new RV $g(X,Y) = Y/X$ involves deriving a distribution.

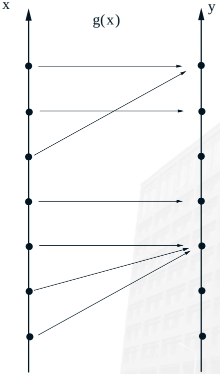
- Don't need PDF for $g(X,Y)$ if only want to compute expected value:

$$\mathbf{E}[g(X,Y)] = \iint g(x,y) f_{X,Y}(x,y)\,dx\,dy$$

Discrete case

- Obtain PMF for each possible value of $Y = g(X)$

$$p_Y(y) = \mathbf{P}(g(X) = y) = \sum_{x:\, g(x)=y} p_X(x)$$



Principal Method

Two-step procedure for the calculation of the PDF of a function $Y = g(X)$ of a continuous RV X

- Calculate the CDF F_Y of Y: $F_Y(y) = \mathbf{P}(Y \le y)$
- Differentiate to get $f_Y(y) = \dfrac{dF_Y}{dy}(y)$

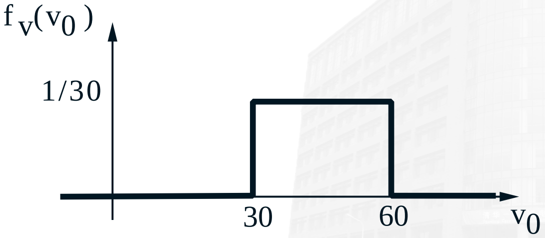
- Example
  - X: uniform on [0,2], Find PDF of $Y = X^3$

$$F_Y(y) = \mathbf{P}(Y \le y) = \mathbf{P}(X^3 \le y) = \mathbf{P}(X \le y^{1/3}) = \frac{1}{2} y^{1/3}$$

$$f_Y(y) = \frac{dF_Y}{dy}(y) = \frac{1}{6y^{2/3}}$$

Example

- Joan is driving from Boston to New York. Her speed is uniformly distributed between 30 and 60 mph. The distance between two cites is 200 miles. What is the distribution of the duration of the trip?
- Let $T(V) = \dfrac{200}{V}$.
- Find $f_T(t)$



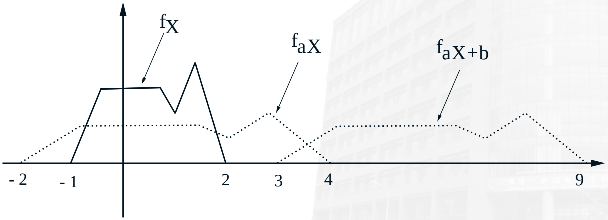
The Linear Case

Let X be a continuous RV with PDF f_X, and let

$$Y = aX + b,$$

where a and b are scalars, with $a \neq 0$. Then

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$



The PDF of Y=aX+b

- To verify this formula, we first calculate the CDF of Y and then differentiate it. For the case where $a > 0$

$$\begin{aligned}
F_Y(y) &= \mathbf{P}(Y \leq y) \\
&= \mathbf{P}(aX + b \leq y) \\
&= \mathbf{P}(X \leq \frac{y-b}{a}) \\
&= F_X(\frac{y-b}{a})
\end{aligned}$$

- We now differentiate this equality and use the chain rule to obtain

$$f_Y(y) = \frac{dF_Y}{dy}(y) = \frac{1}{a} f_X(\frac{y-b}{a})$$

The PDF of Y=aX+b

- If X is normal, then $Y = aX + b$ is also normal.
- Suppose X is normal with mean μ and variance σ^2, then

$$
\begin{aligned}
f_Y(y) &= \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \\
&= \frac{1}{|a|} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ -\left(\frac{y-b}{a} - \mu\right)^2 / 2\sigma^2 \right\} \\
&= \frac{1}{\sqrt{2\pi}|a|\sigma} \exp\left\{ -\frac{(y-b-a\mu)^2}{2a^2\sigma^2} \right\}
\end{aligned}
$$

- Thus, Y is a normal RV with mean $b + a\mu$ and variance $a^2\sigma^2$.

The PDF of a strictly monotonic function

The Monotonic Case

Suppose that $g(\cdot)$ is strictly monotonic and that for some function h and all x in the range of X we have

$$y = g(x) \quad \text{if and only if} \quad x = h(y).$$

Assume that h is differentiable. Then, the PDF of Y in the range where $f_Y(y) > 0$ is given by

$$f_Y(y) = f_X(h(y)) \left| \frac{dh}{dy}(y) \right|.$$

- Follow the two-step process and use the chain rule.
- Cases with g monotonically increasing or decreasing

Example

- Let $Y = g(X) = X^2$, where X is continuous uniform RV on the interval $(0, 1]$.
- Within this interval, g is strictly monotonic, and its inverse is $h(y) = \sqrt{y}$. Thus, for any $y \in (0, 1]$, we have

$$f_X(\sqrt{y}) = 1, \quad \left|\frac{dh}{dy}(y)\right| = \frac{1}{2\sqrt{y}},$$

and

$$f_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & \text{if } y \in (0, 1], \\ 0, & \text{otherwise.} \end{cases}$$

Outline

1. Let X have the normal distribution with mean 0 and variance 1. Also let $Y = g(X)$ where

$$g(t) = \begin{cases} -t, & \text{for } t \leq 0, \\ \sqrt{t}, & \text{otherwise.} \end{cases}$$

Find the probability density function of Y.

2. Let X and Y be independent standard normal RVs. The pair (X, Y) can be described in polar coordinates in terms of RVs $R \geq 0$ and $\Theta \in [0, 2\pi]$, so that

$$X = R\cos\Theta, \quad Y = R\sin\Theta$$

Show that R and Θ are independent. Furthermore,

(a) Find $f_R(r)$.
(b) Find $f_\Theta(\theta)$.
(c) Find $f_{R,\Theta}(r, \theta)$.

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Summary

- Discrete case
- Continuous case
  - Obtain CDF of Y: $F_Y(y) = \mathbf{P}(Y \leq y)$
  - Differentiate to get $f_Y(y) = \dfrac{dF_Y}{dy}(y)$
- The linear case
- The monotonic case

- Readings: Section 4.1
- Assignment: Problem Set 8
- Next lecture: Convolution, covariance and correlation